

ARABICA

*Should Hajar Coffee Co. Automate Its Roast-to-Pack
Production Line?*

Corporate Finance

Submitted for the partial fulfilment of the requirements of
the Corporate Finance module of the Master of Artificial
Intelligence in Business.

Subject: Corporate Finance

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Repository: <https://github.com/samuelalex3737/ARABICA>

Live Application: <https://arabica-a5gq.vercel.app/>

*Topic - AI-Powered Capital Budgeting Application for Specialty Coffee Production
Automation*

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1 Company and Project Background

Hajar Coffee Co. is a realistic fictional specialty coffee roaster based in Alserkal Avenue, Dubai, currently operating a small-batch, largely manual roasting and packing operation. The company supplies a mix of hotel wholesale clients, café/restaurant accounts, and retail bags. Demand from UAE hospitality clients has grown faster than the current artisan setup can reliably supply, creating a capacity constraint that limits the company's ability to win larger wholesale contracts. Management is therefore evaluating whether to invest in automated production equipment to scale output, improve consistency, and secure larger multi-year hotel contracts.

2 Description of the Investment Decision

The decision is a capacity-expansion and automation investment, evaluated across three mutually exclusive alternatives over an 8-year horizon:

- **Alpha (Full Automation):** A complete roast-to-pack line (industrial roaster, automated VFFS packing, NIR quality sensors, automated green-bean handling), enabling the largest wholesale volume and lowest variable cost per kilogram.
- **Beta (Semi-Automation):** A smaller semi-industrial roaster with manual-load, auto-seal packing — a middle-ground capital outlay with moderate throughput.
- **Gamma (Status Quo):** Continuing the current manual, artisan-scale operation with no new capital investment beyond existing working capital.

An AI-powered web application (*ARABICA*) was built to let a non-finance user configure any of the three alternatives and instantly see all required capital budgeting outputs, calculated live in the browser. The application is live at <https://arabica-a5gq.vercel.app/>, with full source code at <https://github.com/samuelalex3737/ARABICA>.

3 Project Assumptions and Data Sources

All figures are in AED. Alpha's capex (AED 4,034,000 equipment + AED 770,000 installation) was built bottom-up from indicative vendor pricing for industrial roasters (Probat, Loring, Giesen), automated packing lines, and UAE freight/commissioning costs, at 1 USD = 3.6725 AED. Beta and Gamma were sized proportionately at smaller equipment tiers. The 11% WACC reflects a blended cost of debt and equity for an SME-scale UAE manufacturer. The 9% tax rate matches the UAE Federal Corporate Tax effective June 2023, applied uniformly for simplicity rather than exempting the first AED 375,000. Straight-line depreciation is used over the 8-year life.

4 Relevant and Irrelevant Cash Flows

Only incremental cash flows attributable to the decision are included in the analysis: the initial equipment and installation outlay, the net working capital injection and its eventual recovery, the incremental after-tax operating cash flows generated by each alternative's revenue and cost structure, and the after-tax salvage value at the end of the project life. Costs that would be incurred regardless of which alternative is chosen — such as general company overhead unrelated to the production line, or any past feasibility or market-research spending already committed before the decision point — are treated as irrelevant and excluded from the cash flow model, consistent with the sunk-cost principle discussed below.

5 Treatment of Sunk Costs, Opportunity Costs, Working Capital, Depreciation, Taxes, and Salvage Value

Sunk costs. Any exploratory costs incurred before this decision are excluded from every alternative's initial outlay, since they would be incurred regardless of which option is chosen and are therefore irrelevant, keeping every reported number directly traceable to the application's calculation engine.

Opportunity costs. Gamma is modelled as continuing to use existing manual equipment rather than assuming its immediate disposal, so no resale proceeds are credited to Alpha or Beta at Year 0 — a deliberately conservative treatment that avoids crediting automation with a speculative resale value for equipment that may retain use as backup capacity.

Working capital. Each alternative requires a net working capital injection at Year 0 (AED 480,000 Alpha, AED 320,000 Beta, AED 200,000 Gamma), fully recovered as a cash inflow in the terminal year.

Depreciation and taxes. Depreciation is non-cash but creates a real tax shield: $OCF = (Revenue - FixedCosts - VariableCosts - Depreciation) \times (1 - T) + Depreciation$, so higher depreciation lowers taxable income and cash tax paid, even though it is added back.

Salvage value. Equipment retains residual value at year 8 (AED 600,000 Alpha, AED 280,000 Beta, AED 0 Gamma). Any gain over book value is taxed at 9% before entering the terminal cash flow.

6 Capital Budgeting Calculations and Results

All thirteen required outputs were computed inside the application for each alternative, using standard formulas: $NPV = \sum_{t=0}^n CF_t / (1 + r)^t$; IRR solved via Newton-Raphson with a bisection fallback; MIRR compounding positive flows and discounting negative flows at the WACC; and the Profitability Index as $PI = 1 + NPV / |CF_0|$. Focusing on the base case for Alpha (Full Automation):

- Initial outlay (CF_0): AED $-5,284,000$ (equipment AED $4,034,000$ + installation AED $770,000$ + working capital AED $480,000$)
- Annual operating cash flow: approximately AED $1,117,455$ per year, derived from AED $3,200,000$ revenue, 42% variable costs, AED $680,000$ fixed costs, and straight-line depreciation
- Terminal cash flow (Year 8): approximately AED $2,143,455$, including the recovered working capital and after-tax salvage value

Metric	Alpha (Full Automation)
Net Present Value (NPV)	AED 935,201
Internal Rate of Return (IRR)	15.45%
Modified IRR (MIRR)	13.28%
Profitability Index (PI)	1.18
Payback Period	4.73 years
Discounted Payback Period	7.02 years
Accounting Rate of Return (ARR)	20.12%
Accounting Break-Even Revenue	AED 2,078,448

Table 1: Alpha (Full Automation) — Base Case Capital Budgeting Results

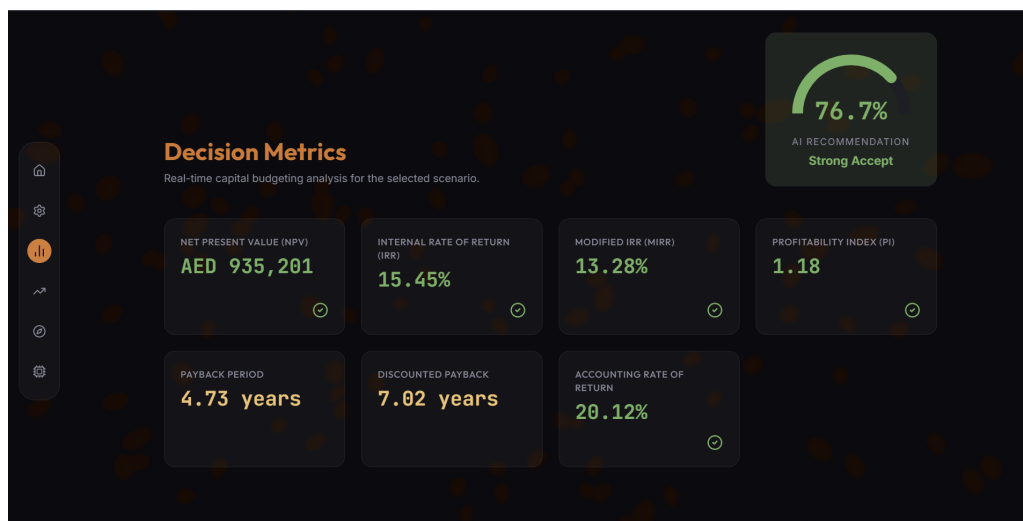


Figure 1: Decision Metrics dashboard for Alpha, showing all required outputs computed live.

Since NPV is positive, IRR (15.45%) exceeds the 11% hurdle rate, and PI exceeds 1.0, all three primary decision rules agree: Alpha creates shareholder value and should be accepted on financial grounds alone.

7 Comparison of Investment Alternatives

Metric	Alpha (Full Auto)	Beta (Semi-Auto)	Gamma (Status Quo)
Initial Investment (AED)	5,284,000	2,220,000	200,000
Annual Revenue (AED)	3,200,000	2,088,000	1,152,000
Variable Cost %	42.00%	48.00%	55.00%
Net Present Value (AED)	935,201	783,582	534,909
Internal Rate of Return	15.45%	19.51%	62.97%
Profitability Index	1.18	1.35	3.67
Payback Period	4.7 yrs	4.2 yrs	1.6 yrs

Table 2: Side-by-side comparison of all three alternatives, taken directly from the live application.

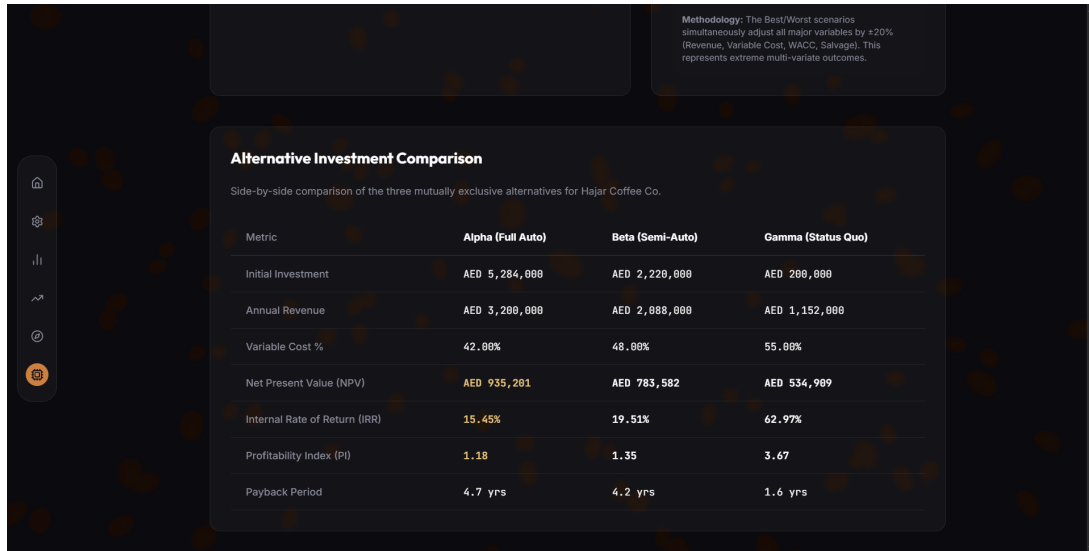


Figure 2: Alternative Investment Comparison as rendered in the application.

All three alternatives clear the 11% hurdle rate — notably, even the manual status quo (Gamma) is financially sound, returning a very high IRR relative to its small investment base. However, IRR and PI are scale-blind: Gamma’s high percentage return is earned on a tiny AED 200,000 base and contributes far less absolute value than Alpha. Since these are mutually exclusive alternatives competing for the same capital budget, **NPV is the appropriate decision criterion**, and Alpha’s AED 935,201 is the largest of the three.

8 Sensitivity and Scenario Analysis

A tornado sensitivity analysis was run on Alpha, varying six key inputs by $\pm 20\%$ individually while holding all others at their base value:

Variable	NPV at -20%	NPV at $+20\%$
Annual Revenue	−AED 803,118	AED 2,673,520
Variable Cost %	−AED 323,582	AED 2,193,984
Project Life	AED 87,542	AED 1,621,605
Initial Equipment Cost	AED 175,110	AED 1,695,292
Discount Rate (WACC)	AED 442,439	AED 1,497,191
Salvage Value	AED 890,077	AED 980,325

Table 3: Tornado sensitivity of Alpha’s NPV (base case: AED 935,201), ranked by impact.

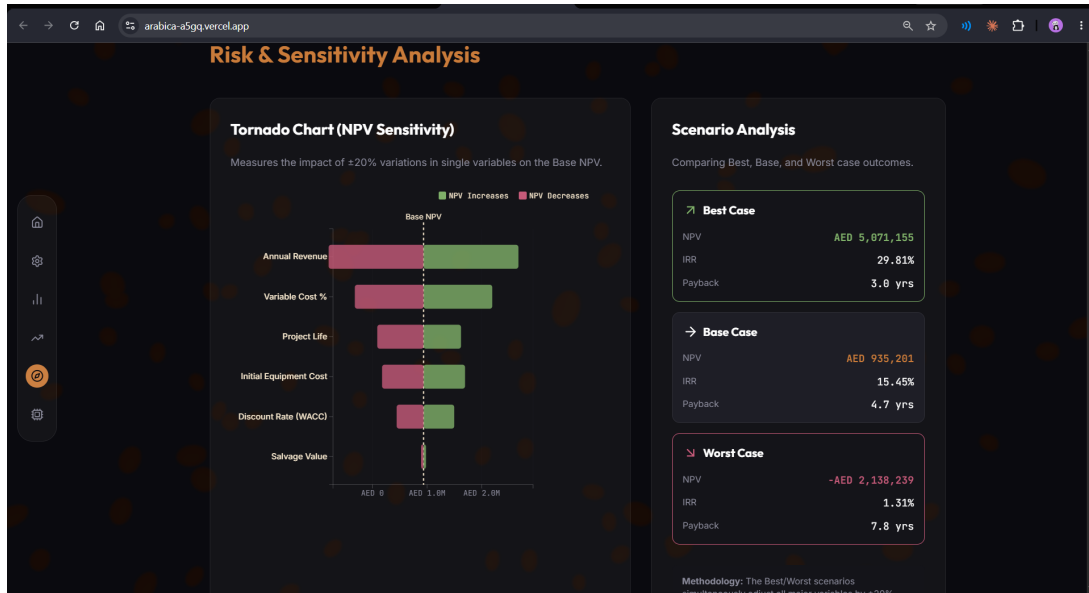


Figure 3: Tornado chart and Best/Base/Worst scenario cards from the application.

Annual Revenue is by far the most critical driver — a 20% shortfall alone turns NPV negative — followed by Variable Cost %. Salvage Value has almost no effect. A combined scenario shifting Revenue, Variable Cost %, WACC, and Salvage by $\pm 20\%$ produced a best case NPV of AED 5,071,155 (IRR 29.81%, payback 3.0 years) and a worst case of –AED 2,138,239 (IRR 1.31%, payback 7.8 years) — confirming the base case is comfortably positive but genuinely sensitive to sustained demand or pricing downturn.

9 Financial and Non-Financial Risks

Financial risks. The dominant risk is revenue shortfall — the sensitivity analysis shows this alone can flip the decision. A secondary risk is cost overrun on imported equipment and installation, given exposure to international freight and GCC import duties. **Non-financial risks** include operational disruption during the transition from manual to automated production (retraining staff, temporary downtime), and supply-chain risk on spare parts for specialised European roasting and packing equipment, which could extend unplanned downtime if a failure occurs.

10 AI-Generated Insights and Student Evaluation

The application uses two distinct AI components, kept separate so quantitative and qualitative analysis do not contaminate each other. First, a **genuinely-trained machine learning model** — a logistic regression classifier built from scratch in TypeScript (no external ML libraries), trained via mini-batch gradient descent on 50,000 Monte Carlo-simulated scenarios with an 80/20 train/test split, achieving approximately 95.7% held-

out accuracy and 0.994 ROC-AUC. For Alpha’s base case it outputs a 76.7% accept-probability. Second, a **Groq-hosted LLM** (Llama 3.1 8B Instant) generates a plain-language narrative, key drivers, and risks from the computed figures, without inventing numbers of its own.

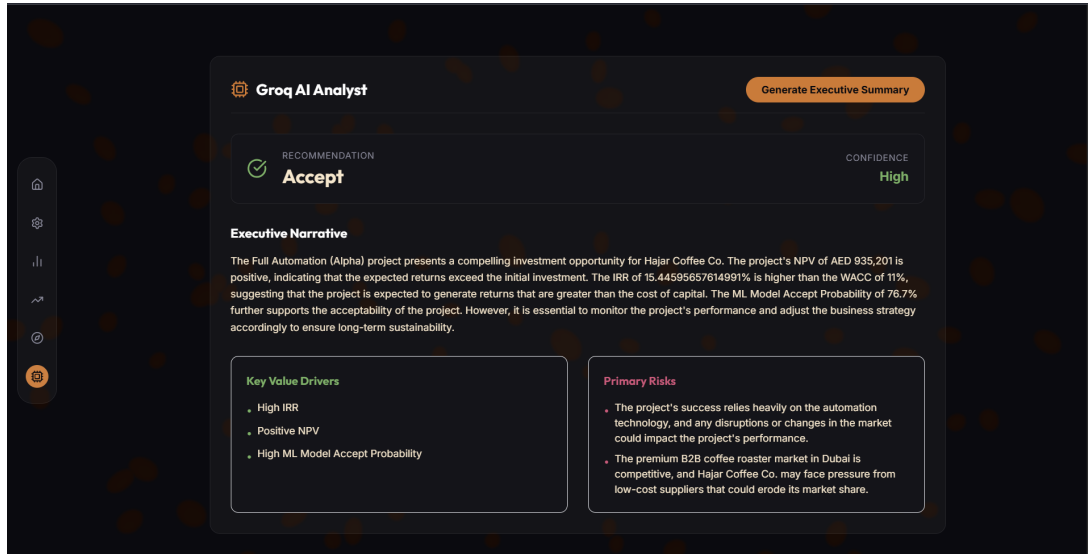


Figure 4: AI Insights panel showing both the trained ML accept-probability model and the Groq narrative analyst.

Student evaluation: The two AI layers are useful for different reasons. The trained classifier is genuinely informative for Alpha and Beta, whose inputs fall well within its training distribution, but it becomes unreliable at the extremes — for Gamma, whose AED 200,000 investment sits far below the model’s original training range, the model initially returned an implausibly low accept-probability despite Gamma having the strongest IRR and PI of all three alternatives. Widening the training distribution improved this, but the episode is a useful, honestly-reported illustration of a linear classifier’s limits outside its training data, rather than something to hide. The Groq narrative layer is fluent and correctly grounded in the real numbers it is given, but it cannot assess qualitative factors such as staff resistance to automation or supplier relationship risk — it is a synthesis tool, not a substitute for management judgement, which is why the final call in this report rests on the student’s own analysis of NPV, sensitivity, and risk, not on either AI output alone.

11 Final Investment Recommendation

Accept Alternative Alpha (Full Automation). Despite requiring the largest capital outlay of the three alternatives (AED 5,284,000), Alpha delivers the highest absolute NPV (AED 935,201), comfortably clears its 11% hurdle rate with a 15.45% IRR, and

remains the value-maximising choice once the scale-distorting effect of IRR/PI on Gamma is accounted for. The investment is justified provided management actively manages the revenue-side risk identified in the sensitivity analysis — principally by securing multi-year wholesale contracts before committing capital, rather than relying on uncontracted demand.

12 Limitations of the Analysis

The model assumes a constant WACC and 9% tax rate over the full 8-year horizon, unlikely given macroeconomic change over that period. Straight-line depreciation simplifies what is often a faster early-year decline in practice. Beta and Gamma's figures were derived proportionately from Alpha's grounded pricing rather than independent vendor quotes, and should be read as directionally reasonable rather than precise. Finally, the ML classifier's output is only as reliable as its Monte Carlo training distributions and, as observed during development, degrades at the extremes of the input space — it is a supporting signal, not a standalone verdict.

13 References

- UAE Ministry of Finance. *Federal Decree-Law No. 47 of 2022 on the Taxation of Corporations and Businesses*.
- Central Bank of the UAE. *Base Rate and EIBOR publications*.
- Probat, Loring, and Giesen: publicly listed industrial coffee roaster specifications and indicative pricing, used as a basis for equipment cost estimation.
- Brealey, R., Myers, S., & Allen, F. *Principles of Corporate Finance*. McGraw-Hill Education.

Declaration of AI Tools Used

- **Claude (Anthropic):** Used for assignment planning, prompt design, independent verification of financial calculations, and critical review of application output and report content.
- **Google Gemini (via Antigravity IDE):** Used as the primary coding agent for building the React application, financial engine, machine learning pipeline, and styling.
- **Groq API (Llama 3.1 8B Instant):** Embedded in the live application to generate the AI Insights narrative, risk summary, and recommendation shown to the end user.

All AI-generated code and content was reviewed, independently verified against hand calculations, and critically evaluated by the student before inclusion in this report.